

Average Reward:

$$\eta(\pi) = \sum_{s,a} p^\pi(s) \pi(a|s) R(s,a)$$

$$\nabla \eta(\theta) = \sum_{s,a} p^\pi(s) \nabla \pi(a|s, \theta) \theta^\pi(s,a)$$

↓
(policy
parameterization)

[Gradient is calculated
in the book by Sutton.
See policy gradient
theorem for continuing
task/average rewards]

[Can we calculate the above?]

Define the natural gradient:

For any state s , the set of $\pi(a|s, \theta)$ is
a manifold. $\rightarrow p(x, \theta) = \pi(x|s, \theta)$

For the manifold

$$g_{ij}(\theta) = \langle \partial_i, \partial_j \rangle = E \left[\frac{\partial}{\partial \theta_i} \ln \pi(a|s, \theta) \frac{\partial}{\partial \theta_j} \ln \pi(a|s, \theta) \right]$$

↑
[averaging over what]

Define $F_s(\theta) \triangleq \left[g_{ij}(\theta) \right]_{\substack{i=1, \dots, n \\ j=1, \dots, n}}$

Natural gradient is:

$$(F_s(\theta))^{-1} \nabla \eta(\theta)$$

We will average over states s .

Steepest direction of descent is

$$F(\theta)^{-1} \nabla \eta(\theta), \leftarrow$$

$$F(\theta) = E_s [F_s(\theta)]$$

Define Compatible Function Approximation

Policy gradient:

$$\nabla \eta(\theta) = \sum_{s,a} p^\pi(s) \nabla \pi(a|s, \theta) Q^\pi(s,a)$$

$f^\pi(s,a;\omega)$ is a compatible function

approximation if

$$\begin{aligned} \nabla_{\theta} \eta(\theta) &= \sum_{s,a} p^\pi(s) \nabla \pi(a|s, \theta) \underbrace{Q^\pi(s,a)}_{f^\pi(s,a;\omega)} \\ &= \sum_{s,a} p^\pi(s) \nabla \pi(a|s, \theta) \underbrace{f^\pi(s,a;\omega)} \end{aligned}$$

Derive sufficient conditions for a COMPATIBLE FUNCTION APPROXIMATION

We don't know $Q^\pi(s,a)$.

Instead we have an approximation $f^\pi(s,a;w)$.

We would like the approximation to minimize:

$$\sum_s \rho^\pi(s) \sum_a \pi(a|s, \theta) [Q^\pi(s,a) - f^\pi(s,a;w)]^2$$

At the minimum the gradient of the above should be 0.

$$\sum_s \rho^\pi(s) \sum_a \pi(a|s, \theta) (Q^\pi(s,a) - f^\pi(s,a;w)) \nabla_w f^\pi(s,a;w) = 0. \quad \text{--- ①}$$

In Sutton's paper, the above condition is the requirement that the value function approximation f_w is compatible with the policy parametrization in the sense that

$$\nabla_w f_w(s,a) = \nabla_\theta \pi(a|s) \frac{1}{\pi(a|s)} \quad \text{--- ②}$$

are shown to result in the true policy gradient.

$$\nabla \eta(\theta) = \sum_s \rho^\pi(s) \sum_a \nabla_\theta \pi(a|s) \theta_\pi(s,a)$$

The exact gradient can be calculated using the approximation $f_w(s,a)$, which satisfies the above two conditions!

PROOF:

$$\sum_s \rho^\pi(s) \sum_a \pi(a|s, \theta) (Q^\pi(s,a) - f^\pi(s,a;w)) \nabla_w f^\pi(s,a;w) = 0. \quad \text{--- ①}$$

$$\Rightarrow \sum_s \rho^\pi(s) \sum_a \pi(a|s, \theta) \theta_\pi(s,a) \nabla_w f^\pi(s,a;w) = \sum_s \rho^\pi(s) \sum_a \pi(a|s, \theta) f^\pi(s,a;w) \nabla_w f^\pi(s,a;w)$$

$$\text{Replace } \nabla_w f^\pi(s,a;w) \text{ by } \nabla_\theta \pi(a|s) \frac{1}{\pi(a|s)}.$$

$$\sum_s \rho^\pi(s) \sum_a \pi(a|s, \theta) \nabla_\theta \pi(a|s) = \sum_s \rho^\pi(s) \sum_a \underbrace{f^\pi(s,a;w)}_{\text{by ②}} \nabla_\theta \pi(a|s)$$

How do we come up with $f_w(s,a)$

Let satisfies

$$\nabla_w f^\pi(s,a;w) = \nabla_\theta \pi(a|s,\theta) \frac{1}{\pi(a|s,\theta)} ?$$

Given the compatible function approximation

$$f^\pi(s, a; \omega) = \psi^\pi(s, a)^T \omega,$$

$$\psi^\pi(s, a) = \nabla_\theta \pi(a|s, \theta) \frac{1}{\pi(a|s, \theta)}$$

calculate ω^* that minimizes the value approximation error.

$$e(\omega, \pi) = \sum_{s, a} p^\pi(s) \pi(a|s, \theta) (f_\omega(s, a) - Q_\pi(s, a))^2$$

$$\sum_{s, a} p^\pi(s) \pi(a|s, \theta) (f_\omega(s, a) - Q_\pi(s, a)) \nabla_\omega f_\omega(s, a) = 0$$

$$\Rightarrow \sum_{s, a} p^\pi(s) \pi(a|s, \theta) (\psi^\pi(s, a)^T \omega) \psi^\pi(s, a)$$

$$= \sum_{s, a} p^\pi(s) \pi(a|s, \theta) Q_\pi(s, a) \underbrace{\nabla_\omega f_\omega(s, a)}_{\psi^\pi(s, a)}$$

$$= \frac{1}{\pi(a|s, \theta)} \nabla_\theta \pi(a|s, \theta)$$

Recall:

$$\nabla \eta(\theta) = \sum_s p^\pi(s) \sum_a \nabla_\theta \pi(a|s) Q_\pi(s, a)$$

$$\Rightarrow \sum_{s, a} p^\pi(s) \pi(a|s, \theta) (\underbrace{\psi^\pi(s, a)^T \omega}_{\frac{1}{\pi(a|s, \theta)} \nabla_\theta \pi(a|s, \theta)}) \psi^\pi(s, a) = \nabla \eta(\theta)$$

$$\frac{1}{\pi(a|s, \theta)} \nabla_\theta \pi(a|s, \theta)$$

$$\omega^* = (F(\theta))^T \nabla \eta(\theta)$$

From the paper:

"Sensible actor-critic frameworks using the approximation $f_w(s,a)$ are forced to use the natural gradient as the weights of a linear function approximation."

"Large action-values have feature vectors with a large inner product with the natural gradient."

GREEDY POLICY IMPROVEMENT

The policy improvement step would choose the greedy action, for any state.

$$\underset{a}{\operatorname{argmax}} f^\pi(s, a; w).$$

Updating θ as $\theta' = \theta + \alpha \tilde{\nabla} \eta(\theta)$,
that is updating θ in the direction of the natural gradient, moves the policy locally toward the best (greedy) action.

$$\pi(a|s, \theta') = \pi(a|s, \theta + \alpha \tilde{\nabla} \eta(\theta))$$

$$\approx \pi(a|s, \theta) + \underbrace{\nabla_{\theta} \pi(a|s, \theta)^T}_{\text{Write in terms of the compatible function approx. } f^\pi(s, a; w).}$$

Set w to ?

$$\left| \begin{array}{l} f^\pi(s, a; w) \\ = \frac{\nabla_{\theta} \pi(a|s, \theta)^T}{\pi(a|s, \theta)} w \end{array} \right| \left| \begin{array}{l} \nabla f^\pi(s, a; w) \\ = \frac{\nabla_{\theta} \pi(a|s, \theta)}{\pi(a|s, \theta)} \\ \Rightarrow \nabla_{\theta} \pi(a|s, \theta) \\ = \pi(a|s, \theta) \nabla f^\pi(s, a; w) \end{array} \right|$$

$$= \pi(a|s, \theta) + \alpha \pi(a|s, \theta) f^\pi(s, a; w)$$

$$= \pi(a|s, \theta) (1 + \alpha f^\pi(s, a; w))$$

TRUST REGION POLICY (TRPD) OPTIMISATION

Guarantee policy improvement with
non-biased step sizes.

Define the expected sum discounted
reward as

$$q(\pi) = E_{s_0, a_0, s_1, a_1, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t(s_t) \right]$$

Note that:

$$q(\pi) = E \left[\sum_{t=0}^{\infty} \gamma^t G_t | S_0 \right]$$

Where $s_0 \sim p_0(s_0)$, $a_t \sim \pi(a_t | s_t)$
 $s_{t+1} \sim p(s_{t+1} = s_{t+1} | s_t = s_t, a_t = a_t)$

Show that:

$$q(\tilde{\pi}) = q(\pi) + E_{s_0, a_0, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right]$$

Where, the advantage

$$A_{\pi}(s_t, a_t) = Q_{\pi}(s_t, a_t) - V_{\pi}(s_t)$$

$$E_{s_0, a_0, s_1, a_1, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right] =$$

$$\sum_{(s_0, a_0, \dots)} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right] \pi(s_0, a_0, s_1, a_1, \dots)$$

$$= \sum_{(s_0, a_0, \dots)} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right] \left(p(s_0) \pi(a_0 | s_0) p(s_1 | s_0, a_0) \pi(a_1 | s_1) \right. \\ \left. \times p(s_2 | s_1, a_1) \pi(a_2 | s_2) \right. \\ \left. \vdots \right)$$

keeping the above in mind, let's rewrite

$$E_{s_0, a_0, s_1, a_1, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(s_t, a_t) \right]$$

$$= E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (Q_{\pi}(s_t, a_t) - V_{\pi}(s_t)) \right]$$

Note that, for (s_t, a_t) , we have:

$$Q_{\pi}(s_t, a_t) = E_{\pi} [R_{t+1} + \gamma V_{\pi}(s_{t+1}) | s_t = s_t, a_t = a_t]$$

$$\therefore Q_{\pi}(s_t, a_t) = E_{\pi} [R_{t+1} + \gamma V_{\pi}(s_{t+1}) | s_t, a_t]$$

$$\therefore E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (Q_{\pi}(s_t, a_t) - V_{\pi}(s_t)) \right] = E_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (E_{\pi} [R_{t+1} + \gamma V_{\pi}(s_{t+1}) | s_t, a_t] - V_{\pi}(s_t)) \right]$$

$$= E_{s_0, a_0, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t (R_{t+1} + \gamma V_{\pi}(s_{t+1}) - V_{\pi}(s_t)) \right]$$

$$= E_{s_0, a_0, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t \begin{matrix} R_{t+1} + V_{\pi}(s_t) - V_{\pi}(s_0) \\ + \gamma^2 V_{\pi}(s_2) - \gamma V_{\pi}(s_1) \\ \vdots \end{matrix} \right]$$

$$= E_{s_0, a_0, \dots, \pi} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \right]$$

$$- E_{s_0, a_0, \dots, \pi} [V_{\pi}(s_0)]$$

$$= q(\tilde{\pi}) - q(\pi)$$

Show that:

$$\eta(\tilde{\pi}) = \eta(\pi) + \underbrace{\sum_s p_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s,a)},$$

Where

$$p_{\tilde{\pi}}(s) = p_{\pi}[S_0=s] + \gamma p_{\pi}[S_1=s] + \gamma^2 p_{\pi}[S_2=s] + \dots$$

We have that

$$\eta(\tilde{\pi}) = \eta(\pi) + E_{S_0, a_0, \dots \sim \tilde{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A_{\pi}(S_t, A_t) \right]$$

$$= \eta(\pi) + \sum_{t=0}^{\infty} \gamma^t E_{S_t, a_t, \dots \sim \tilde{\pi}} [A_{\pi}(S_t, A_t)]$$

$$= \eta(\pi) + \sum_{t=0}^{\infty} \gamma^t \sum_{s,a} A_{\pi}(s,a) p_{\tilde{\pi}}[S_t=s, A_t=a]$$

$$= \eta(\pi) + \sum_{t=0}^{\infty} \gamma^t \sum_{s,a} A_{\pi}(s,a) \underbrace{p_{\tilde{\pi}}[A_t=a|S_t=s]}_{\downarrow} \underbrace{p_{\tilde{\pi}}[S_t=s]}$$

$$= \eta(\pi) + \sum_{t=0}^{\infty} \gamma^t \sum_{s,a} p_{\tilde{\pi}}[S_t=s] \underbrace{\tilde{\pi}(a|s)}_{\downarrow} A_{\pi}(s,a)$$

$$= \eta(\pi) + \underbrace{\sum_{t=0}^{\infty} \gamma^t \sum_s p_{\tilde{\pi}}[S_t=s]}_{\downarrow} \underbrace{\sum_a \tilde{\pi}(a|s) A_{\pi}(s,a)}_{\phi}$$

$$= \eta(\pi) + \sum_s \underbrace{p_{\tilde{\pi}}(s)}_{\downarrow} \underbrace{\sum_a \tilde{\pi}(a|s) A_{\pi}(s,a)}_{\phi}$$

We showed:

$$\eta(\tilde{\pi}) = \eta(\pi) + \sum_s p_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s,a)$$

Suppose $\tilde{\pi}$ is a greedy improvement,
given $A_{\pi}(s,a)$.

From exact policy iteration, we know that:

$$\eta(\tilde{\pi}) \stackrel{?}{\geq} \eta(\pi)$$

The greedy policy is a deterministic/stochastic
policy.

For the greedy action in state s :

$$A_{\pi}(s,a) \geq 0.$$

Why not simply pick the greedy policy?

Beyond this are a few slides
on TRPO, which may benefit
those who want to read the paper.
They are not included in the
portion for the end sem exam.

To find the best next policy $\tilde{\pi}$ is equivalent to solving the optimisation problem:

$$\underset{\tilde{\pi}}{\text{maximize}} \quad \eta(\pi) + \sum_s P_{\tilde{\pi}}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s,a)$$

The objective $\eta(\tilde{\pi})$ is a function of $P_{\tilde{\pi}}(s)$.

A workaround is to instead optimise a first-order approximation $L_{\pi}(\tilde{\pi})$ of $\eta(\tilde{\pi})$.

A first order approximation at π must satisfy

$$\left. \begin{aligned} L_{\pi}(\pi) &= \eta(\pi), \\ \nabla L_{\pi}(\pi) &= \nabla \eta(\pi) \end{aligned} \right| \begin{array}{l} \tilde{\pi} \text{ is} \\ \text{close enough} \\ \text{to } \pi. \end{array}$$

If we are considering policies parametrised by θ ,

$$\eta(\theta + \Delta\theta) \approx \eta(\theta) + \nabla \eta(\theta)^T \Delta\theta$$

We require

$$L_{\pi}(\theta) = \eta(\theta)$$

$$\nabla L_{\pi}(\theta') \Big|_{\theta'=\theta} = \nabla \eta(\theta') \Big|_{\theta'=\theta}.$$

Consider the first-order approximation:

$$L_{\pi}(\tilde{\pi}) = v(\pi) + \sum_s p_{\pi}(s) \sum_a \tilde{\pi}(a|s) A_{\pi}(s, a)$$

How large can the step size α be,
given the current policy $\pi(\theta)$?

Since we don't know how large the step size can be:

The paper proposes:

an algorithm that optimizes $L_{\pi}(\tilde{\pi})$,
while penalizing "a large step."

The algorithm results in a sequence
of improving policies!

They derive the bound:

$$J(\tilde{\pi}) \geq L_{\pi}(\tilde{\pi}) - C D_{KL}^{\max}(\pi, \tilde{\pi}),$$

$$C = \frac{4EV}{(1-\gamma)^2}, \quad E = \max_{s,a} |A_{\pi}(s,a)|,$$

$$D_{KL}^{\max} = \max_s D_{KL}(\pi(\cdot|s), \tilde{\pi}(\cdot|s))$$

The bound motivates the algorithm:

for i in $(0, \dots)$

$$\pi_{i+1} = \arg \max_{\tilde{\pi}} L_{\pi_i}(\tilde{\pi}) - C D_{KL}^{\max}(\pi_i, \tilde{\pi})$$

end.

What does
this achieve?

Note: We are still assuming exact evaluation of A_{π} . However, we are not restricted to improvements that are deterministic policies.

Show that the above results in a sequence of improvements, that is

$$J(\pi_{i+1}) \geq J(\pi_i), \quad i = 0, 1, \dots$$

Write both in terms of L_{π} .

$$L_{\pi_i}(\pi_i) = J(\pi_i)$$

$$J(\pi_{i+1}) \geq L_{\pi_i}(\pi_{i+1}) - C D_{KL}^{\max}(\pi_i, \pi_{i+1}),$$

$$J(\pi_{i+1}) - J(\pi_i) \geq \underbrace{L_{\pi_i}(\pi_{i+1})}_{L_{\pi_i}(\pi_i)} - C D_{KL}^{\max}(\pi_i, \pi_{i+1}) - L_{\pi_i}(\pi_i)$$

Recall the step

$$\pi_{i+1} = \arg \max_{\tilde{\pi}} L_{\pi_i}(\tilde{\pi}) - C D_{KL}^{\max}(\pi_i, \tilde{\pi})$$

For policies parametrized by θ , we solve

$$\underset{\theta}{\text{maximize}} \quad (L_{\theta_{\text{old}}}(\theta) - \epsilon \overset{\text{max}}{D_{\text{KL}}}(\theta_{\text{old}}, \theta))$$

Two issues in practice:

(1) Penalizing large deviations can result in small step sizes.

(2) Calculating the max in $\overset{\text{max}}{D_{\text{KL}}}$.

(1) is addressed by using a trust-region constraint instead of a penalty.

TRPO solves:

$$\underset{\theta}{\text{Maximize}} \quad L_{\theta_{\text{old}}}(\theta)$$

$$\text{subject to } \underbrace{\bar{D}_{\text{KL}}(\theta_{\text{old}}, \theta)}_{\substack{\downarrow \\ \text{average, using } P_{\theta_{\text{old}}}}} \leq \delta.$$

In practice, expectations in the objective & constraint are replaced by sample estimator. [Read Section 5, "Sample-Based Estimation of the Objective and Constraint"]